

SUKHAYA, Russian Federation

WMO: 307260

Lat: 52.552N

Lon: 107.106E

Elev: 461

StdP: 95.91

Time Zone: 8.00 (E08)

Period: 86-06

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-28.8	-26.0	-31.6	0.2	-28.5	-28.7	0.3	-25.9	10.1	-16.5	9.2	-16.2	2.5	140	0.739

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.3	21.1	17.7	19.9	16.6	18.8	15.9	18.2	20.5	17.2	19.3	16.3	18.3	3.9	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.2	13.0	19.9	16.3	12.2	18.8	15.3	11.5	17.8	53.8	20.6	50.4	19.5	47.6	18.4	21.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.2	9.1	8.0	DB	-30.6	26.1	3.7	2.3	-33.2	27.8	-35.4	29.1	-37.5	30.4	-40.2	32.1	(4)
(5)			WB	-30.8	19.4	3.2	1.2	-33.1	20.3	-35.0	21.0	-36.8	21.7	-39.2	22.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-0.3	-16.5	-16.8	-10.1	-1.4	5.3	10.7	14.7	14.4	8.7	2.5	-5.5	-10.4	(6)	
	DBStd	11.61	4.94	5.56	5.59	3.46	3.30	2.71	2.57	2.29	3.01	3.33	4.43	3.87	(7)	
	HDD10.0	4106	820	751	624	342	151	24	1	1	59	233	467	633	(8)	
	HDD18.3	6799	1078	984	882	592	406	230	117	123	288	492	717	891	(9)	
	CDD10.0	352	0	0	0	0	4	44	146	137	21	1	0	0	(10)	
	CDD18.3	4	0	0	0	0	0	0	3	2	0	0	0	0	(11)	
	CDH23.3	6	0	0	0	0	0	0	4	2	0	0	0	0	(12)	
	CDH26.7	1	0	0	0	0	0	0	1	0	0	0	0	0	(13)	
(14) Wind	WSAvg	3.6	3.3	3.0	3.2	3.4	3.5	3.4	3.2	3.2	3.8	4.0	4.4	4.3	(14)	
(15) Precipitation	PrecAvg	356	12	8	7	14	27	52	74	73	39	16	17	16	(15)	
	PrecMax	436	47	14	23	30	51	109	147	123	85	38	31	46	(16)	
	PrecMin	280	3	1	1	4	7	21	39	32	14	2	7	5	(17)	
	PrecStd	43	8	3	4	6	11	21	26	28	17	8	5	8	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-1.3	0.0	4.8	9.2	17.9	20.4	23.2	22.4	18.0	12.9	5.6	0.4	(19)	
		MCWB	-3.0	-2.4	1.1	4.9	11.0	14.8	19.0	18.3	14.7	8.7	2.0	-1.8	(20)	
	2%	DB	-3.8	-4.6	2.1	6.7	14.8	18.1	21.5	20.8	16.0	10.6	3.5	-1.4	(21)	
		MCWB	-5.4	-6.1	-0.4	2.8	9.8	13.8	18.2	17.6	13.4	7.2	0.7	-3.4	(22)	
	5%	DB	-6.8	-6.4	0.4	5.1	12.7	16.7	20.4	19.6	14.8	9.0	2.1	-3.1	(23)	
		MCWB	-8.1	-7.6	-1.7	2.0	8.5	13.1	17.3	16.7	12.4	6.2	-0.4	-4.9	(24)	
	10%	DB	-8.8	-8.3	-1.3	3.8	10.9	15.4	19.3	18.5	13.5	7.7	0.6	-4.6	(25)	
		MCWB	-9.7	-9.3	-3.0	1.0	7.2	12.4	16.5	15.9	11.2	5.2	-1.6	-6.2	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-2.8	-2.2	1.6	5.2	11.7	16.3	20.0	19.3	15.4	9.1	2.6	-1.7	(27)	
		MCDB	-1.5	0.0	4.2	9.0	15.4	19.0	22.3	21.4	17.0	12.1	4.6	0.1	(28)	
	2%	WB	-5.3	-6.0	-0.1	3.1	10.3	14.7	18.7	18.1	13.7	7.8	1.0	-3.3	(29)	
		MCDB	-3.5	-4.6	1.8	5.9	14.2	16.9	21.0	20.3	15.5	10.1	3.2	-1.6	(30)	
	5%	WB	-8.0	-7.6	-1.6	2.2	8.8	13.7	17.6	17.1	12.7	6.5	-0.2	-4.8	(31)	
		MCDB	-6.9	-6.5	0.2	4.9	12.3	15.9	19.8	19.1	14.5	8.7	1.9	-3.3	(32)	
	10%	WB	-9.7	-9.3	-3.1	1.2	7.5	12.7	16.8	16.2	11.6	5.4	-1.5	-6.1	(33)	
		MCDB	-8.7	-8.3	-1.4	3.5	10.6	15.0	18.9	18.2	13.3	7.4	0.5	-4.7	(34)	
(35) Mean Daily Temperature Range	MDBR		8.0	8.7	9.0	7.2	7.8	7.7	6.3	6.2	6.6	7.1	6.5	6.9	(35)	
	5% DB	MCDBR	8.6	10.2	9.1	8.8	11.4	9.5	8.2	8.1	7.5	8.5	7.7	7.9	(36)	
		MCWBR	7.6	9.0	7.1	6.2	7.2	6.2	5.4	5.5	5.5	6.3	6.3	7.0	(37)	
	5% WB	MCDBR	8.5	9.9	8.7	7.9	10.3	8.2	7.1	7.3	6.8	7.8	7.5	7.9	(38)	
		MCWBR	7.6	8.8	7.0	5.9	7.2	5.9	5.3	5.4	5.3	6.2	6.4	7.0	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.266	0.292	0.322	0.367	0.427	0.444	0.435	0.387	0.337	0.310	0.283	0.266	(40)	
	taud		2.102	2.090	2.146	2.268	2.131	2.197	2.267	2.388	2.476	2.427	2.186	2.072	(41)	
	Ebn at Noon		706	802	858	865	823	812	813	838	847	795	681	622	(42)	
	Edh at Noon		72	100	118	120	145	137	126	106	86	72	66	62	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.15	2.18	3.60	4.83	5.75	6.09	5.64	4.75	3.52	2.16	1.14	0.78	(44)	
	RadStd		0.07	0.12	0.17	0.25	0.39	0.38	0.35	0.21	0.30	0.17	0.09	0.06	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (15 neighbors)	N/A	N/A	-1.78	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page